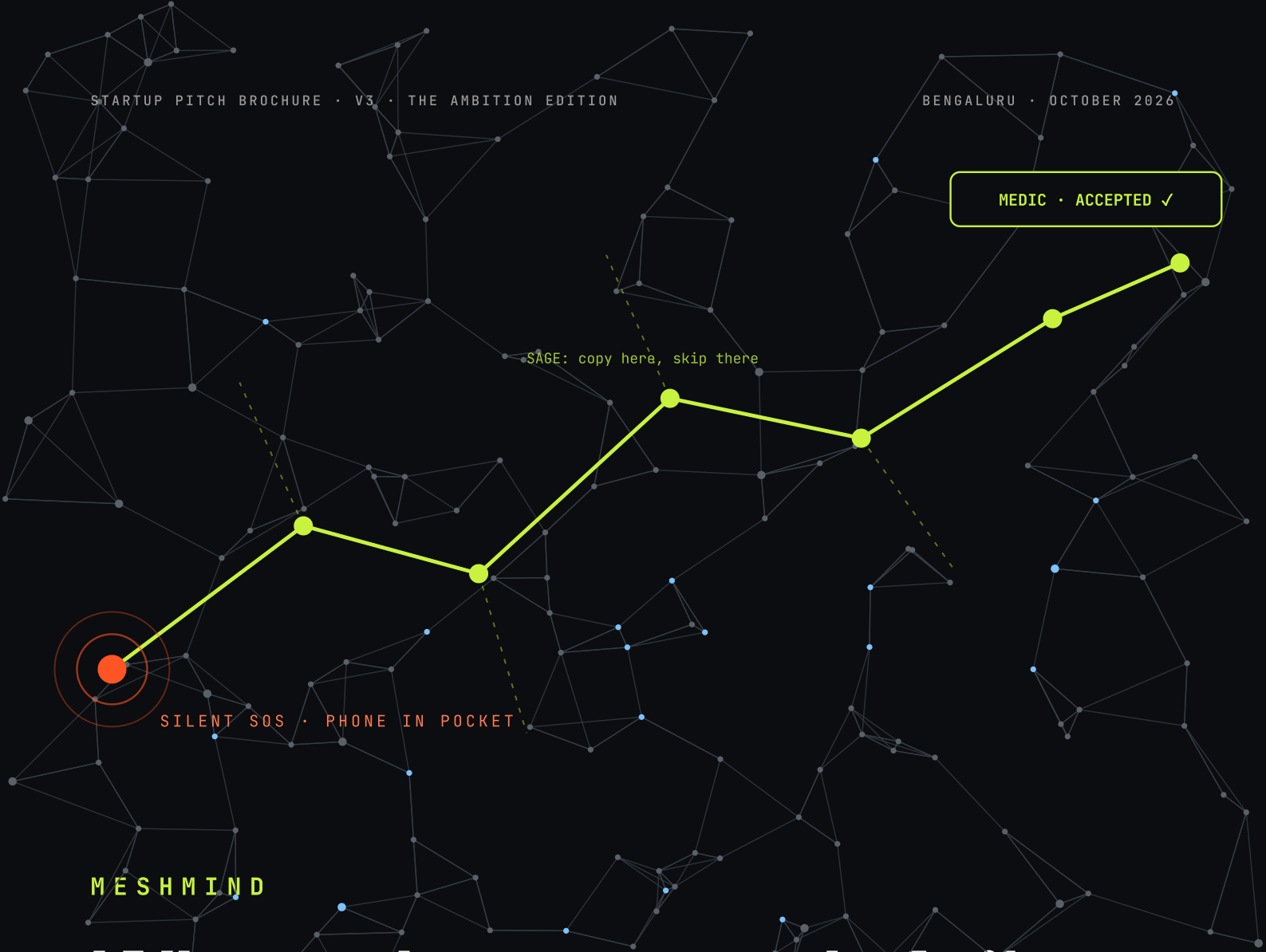


MESHMIND

When the network fails, *the crowd* becomes the network — *and the network starts thinking.*

MeshMind turns the phones already in a crowd into a self-organising, self-healing emergency nervous system. It moves the right information to the right person, senses what the venue cannot see, and keeps the response coordinated when every other network has failed.

251/258
TWO-HOP DELIVERIES ON 3 REAL PHONES · 0 DUPES

145ms
SOS THROUGH A 1 MB TRANSFER

1.0M
ORACLE-LABELLED ROUTING DECISIONS, 11 REAL TRACES

0.81
SAGE AUC AT SPOTTING THE COPIES THAT MATTER (VS 0.56)

00 WHAT CHANGED SINCE V2

In v2 it was a promise.
In v3 it is a system with evidence.

v2 said: "Next proof: true three-phone A → B → C multi-hop" and "AI routing advantage: target only, until benchmarked." Both have moved. This edition raises the ambition — and keeps the same discipline about what is proven.

PROVEN ON PHONES

Three phones, two hops, three phone makers

OnePlus 12R → Realme RMX5108 → Galaxy S24 Ultra over Nearby Connections. A two-hour, 342-attempt physical session: **251 of 258** measured two-hop deliveries succeeded, zero duplicates; an airplane-mode phone fetched the web through two others **30/30** times. Topology was software-enforced (peer filter), not RF-isolated — and we say so.

PROVEN IN THE LAB

SAGE v1 learned what flooding cannot

11 public human-contact traces, 145 independent episodes, 1,001,442 routing decisions labelled by a temporal oracle. On held-out events SAGE identifies which copies actually improve delivery with AUC 0.81; every hand-written rule we tested scores 0.53–0.58. In routing it beat flooding in all 17 held-out stress scenarios.

11
PUBLIC CONTACT TRACES · 4 → 11

800×
MORE LABELLED DECISIONS THAN THE V0 PILOT

15k
PARAMETERS · 75 KB ONNX MODEL

−28%
AIRTIME AT EQUAL DELIVERY VS FLOODING (HELD-OUT SIM)

The most important lab finding was *a failure.*

Our first SAGE pilot "lost" to flooding: 3.3% vs 5.2% delivery. Instead of tuning the model, we measured the benchmark. Even a router with perfect knowledge of the future could only reach 5–12% in that setup — flooding was already at the ceiling. No model could have won. We rebuilt the benchmark so that a perfect router reaches 33–69%, added delivery-ACK gossip, shared radio airtime and real crowd traces, then retrained.

The lesson is the moat: we test whether a test can be won before we celebrate or panic about a result. Investors and organisers get numbers with their ceilings attached.

NEW IN V3 — THE AMBITION

Autonomic Mesh
SAGE runs seven network decisions, not one

Mesh Twin
Forecasts where the event goes blind next

Crowd Pulse
Phones as a consented sensor array

Silent SOS
Help without typing or speaking

Swarm Relays
Idle phones heal the mesh where it matters

Last-Seen Trail
Find a lost child without cameras

Federated SAGE
Every event trains the next one

One rule above all
No model ever touches an SOS

Phone evidence: session physical_20261006_01 logs (A-fbd6 → B-e84d → C-507b). Lab evidence: research/sage v1, held-out episodes never used for training or tuning.

01 THE PITCH IN 90 SECONDS

20,000 phones. One metre apart. *Now they think together.*

A dense crowd is the largest computer most people will ever stand inside: thousands of radios, accelerometers, microphones, barometers and batteries within arm's reach. Today each phone fights alone for a congested tower. When the tower fails, the event goes blind — exactly when someone needs help and someone else needs to take responsibility for helping.

PRODUCT DEFINITION · V3

MeshMind is the *nervous system* of a crowd: a mesh that routes itself, phones that sense together, and a response loop that does not stop at "delivered" — it stops at "resolved".

01 · NERVES

The mesh

Offline transport, store-and-forward, priority pre-emption, Internet through any consenting phone.

02 · REFLEXES

SAGE autonomic routing

The network decides who carries what, what goes first, what waits — and heals its own dead zones.

03 · SENSES

Crowd Pulse

Consented on-device sensing: density trends, motion coherence, falls, floors — evidence, never verdicts.

04 · JUDGEMENT

The response loop

Living Incidents, the one right question, a forecast of where the event is about to go blind.

What MeshMind will never be

- A system where a model can demote, delay or suppress an explicit SOS — tested adversarially
- A crush-prediction product. Crowd sensing is evidence for trained humans
- A certified life-safety system — it is decision support for trained teams
- An Internet proxy: gateways run allowlisted requests only

Why staff first, attendees second

30 equipped stewards make a backbone on day one; attendees opt in through Event Mode and extend its reach. Every attendee phone that joins makes the network denser, the senses sharper and the model smarter — the crowd is both the user and the infrastructure.

"Nerves → reflexes → senses → judgement" is the build order. Each layer ships only when the layer below is measured.

02 THE PROBLEM

In a crush, *nobody can type.* In a blackout, *nobody can see.*

Dense events fail in two ways at once. Communication collapses — and the physical state of the crowd becomes hardest to see. Most mesh messengers assume a human typing a message to a friend. In a crush, people cannot raise their arms; people pushing from behind have no idea anything is wrong; the control room sees a wall of silence and reads it as calm.

HATHRAS · JUL 2024

121

DEATHS · ~250K AT A GATHERING PERMITTED FOR 80K

MAHA KUMBH · JAN 2025

30

DEATHS · 60 INJURED IN THE OFFICIAL COUNT

BENGALURU · JUN 2025

11

DEATHS · 47 INJURED NEAR CHINNASWAMY STADIUM

Where today's tools break

TODAY	FAILURE MODE	MESHMIND V3
Every phone reaches alone for the tower	Congestion → apps stall	**Cooperative local overlay that routes itself**
Messages assume a path exists now	A disconnect becomes a failure	**Packets persist and move when contact returns**
Bulk media and SOS compete blindly	Critical data waits behind megabytes	**Receiver-confirmed backpressure; SOS bypass**
Routing floods every neighbour	Battery and airtime burn on useless copies	**SAGE copies where it measurably helps**
Silence looks like "all clear"	The quietest zone is the most dangerous	**Observability map: stale = unknown, never green**
Coverage gaps are bad luck	A zone drops off the map	**Mesh Twin forecasts the gap; a steward pre-connects it**
SOS needs typing or speaking	Trapped people cannot do either	**Silent SOS: patterns, falls and stillness — opt-in**

CORE INSIGHT

The scarce resource is not bandwidth. It is the brief opportunity to move useful information before the topology — or the crowd — changes again. Spending it well is a learning problem.

Where it matters first

College fests	staff pilots, ideal field trials
Concerts · stadiums	SOS, friend rendezvous, gateway relay
Melas · pilgrimages	lost persons, multilingual SOS, crowd evidence
Floods · blackouts	store-and-forward civic mesh

Sources: Reuters reporting on Hathras (2024), Maha Kumbh (2025) and Bengaluru (2025). MeshMind adds a participant-side layer; it does not replace professional crowd control.

03 THE BIG IDEA

Not an app.
A nervous system for crowds.

Five layers, each with one job, each shipping only when the layer beneath it is measured. The safety envelope wraps all of them: the explicit SOS path is deterministic and never passes through a model.

05 · Judgement — the response loop

BUILDING

Living Incidents that survive network splits · Ask the Right Person · Mesh Twin forecasts · Pre-connect a Zone · organiser copilot on approved templates

04 · Memory — the incident op-log

BUILDING

Signed, hash-chained, causally merged operation log; conflicts preserved for humans; the event's Black Box for inquiries and insurers

03 · Senses — Crowd Pulse

RESEARCH

On-device features from BLE scans, IMU, barometer and sound level — density trends, motion coherence, falls, floors. Raw signals never leave the phone

02 · Reflexes — SAGE, the autonomic mesh

PROVEN IN SIM

Learned relay value, scheduling and eviction today; gateway forecasting, radio duty-cycling and topology sculpting next — all beneath fixed safety rules

01 · Nerves — the mesh

LIVE ON PHONES

Nearby Connections transport · store-and-forward · ACKs · chunking with receiver-confirmed backpressure · SOS pre-emption · Internet via consenting gateway · 3-phone multi-hop

SAFETY ENVELOPE

AI proposes. Deterministic rules dispose.

- SOS bypasses every model
- Models only rank ordinary traffic
- Missing / corrupt model → hand-written fallback
- Shadow mode before authority
- Humans approve every physical movement

THE FLYWHEEL

More phones → denser mesh → more consented telemetry → better-calibrated simulator → better SAGE → more delivered per joule → more reasons to join.

THE CONSTRAINT

Big compute in the lab, small dependable models on the phone. The routing model is 15k parameters; inference is a fraction of a millisecond on a laptop CPU.

THE PROMISE

Every claim carries a status: live on phones, proven in simulation, building, next, research or vision. Nothing moves left without evidence.

Status legend used throughout: LIVE (measured on phones) · PROVEN IN SIM (held-out simulation) · BUILDING (30-day plan) · NEXT (90 days) · RESEARCH (open question) · VISION (12–24 months).

04 EVIDENCE

The core is not a mockup.
It runs on three real phones.

Measured on OnePlus 12R, Realme RMX5108 and Galaxy S24 Ultra, Android 16, over Nearby Connections (P2P_CLUSTER, bearer chosen by Nearby). Prototype measurements, not population benchmarks.

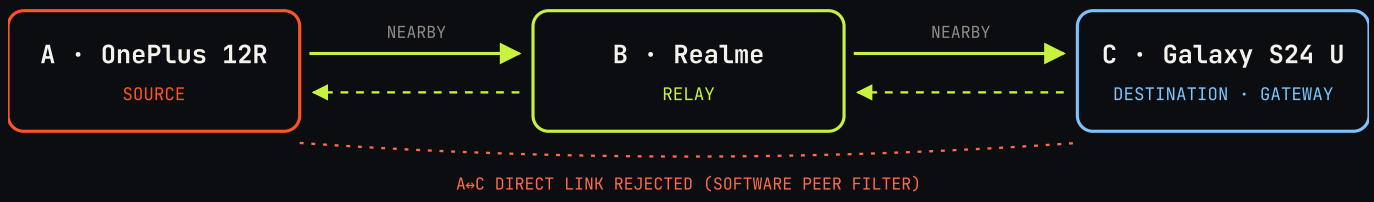
238ms
TEXT ROUND TRIP, 2 HOPS
P50 · 49/50 DELIVERED

424ms
AIRPLANE-MODE WEB FETCH
2 HOPS · 30/30 · P50

440ms
SOS THROUGH A 1 MIB TRANSFER,
2 HOPS · 30/30 · P50

40/40
STORE-AND-FORWARD ACROSS
5–30 S OUTAGES

VALIDATED · THREE-PHONE APPLICATION MULTI-HOP



TEST	OBSERVED RESULT	STATUS
A → B → C text	49/50 delivered over two hops; p50 238 ms, p95 444 ms round trip; the 1 failure was the harness losing ADB	VALIDATED · SW TOPOLOGY
Offline web via 2-hop gateway	A in airplane mode, no validated Internet: 30/30 HTTP 200 from example.com via B → gateway C; p50 424 ms, p95 524 ms	VALIDATED · SW TOPOLOGY
SOS under 128 KiB–1 MiB load	88/90 delivered over two hops with 8–64 chunks queued; every delivered SOS beat the bulk transfer; bulk resumed and SHA-256 matched	VALIDATED · SW TOPOLOGY
Store-and-forward	40/40 delivered after the destination app was stopped for 5, 10, 20 or 30 s; the relay held the packet	VALIDATED
Relay crash recovery	Relay app killed with a packet queued; on restart it reloaded the queue from disk and forwarded it: 9/10, p50 21.7 s	VALIDATED
Cold discovery of a 3-phone chain	5/8 established; p50 70 s from first start until A learns C via B — too slow for an event	WEAKNESS FOUND
Physical RF-isolated multi-hop	Needs separated rooms / attenuation, not a software filter	NOT YET

Session physical_20261006_01, 7 Oct 2026, 00:01–01:56: 342 attempts (258 measured, 30 warm-up, 54 smoke/mini), every failure retained. Topology A↔B↔C enforced by the app's Nearby peer filter; one room. RTTs use source-device timestamps.

04 THE MOST VALUABLE BUG WE FOUND

"Sent" is not "delivered."
We measured the difference.

Nearby's send-completion callback is not an application-level delivery signal. Chunks were reported as sent 1.3–2.9 s before the receiver had them; a large transfer silently piled up inside the transport and the first SOS behind it waited 2,374 ms.

SOS LATENCY DURING A BULK TRANSFER

Before: transport-level "sent"

2,374 ms

After: receiver-confirmed backpressure

145 ms

16×

FASTER CRITICAL DELIVERY UNDER LOAD

THE FIX

Every chunk gets an application-level confirmation. A sender allows at most two unconfirmed chunks per link — urgent packets bypass that window.

```
sender.window = 2   // unconfirmed chunks / link
on chunk_ack(id) → window.release(id)
on enqueue(p):
  if p.priority == CRITICAL
      send_now(p)      // bypass window
  else wait_for(window)
```

FIX 01

Connection recovery

STATUS_ENDPOINT_IO_ERROR could strand two devices. Retry now backs off with jitter.

FIX 02

Lifecycle

Leaving the screen could stop a transfer. Sending now continues independently, with a regression test.

FIX 03

Three phone makers

OnePlus, Realme and Samsung interoperate on one mesh — the mixed-fleet reality of an Indian crowd.

> This is the moat in miniature: competitors can copy a feature list. They cannot copy months of measured radio behaviour on real, mixed Android devices — or a simulator calibrated against it.

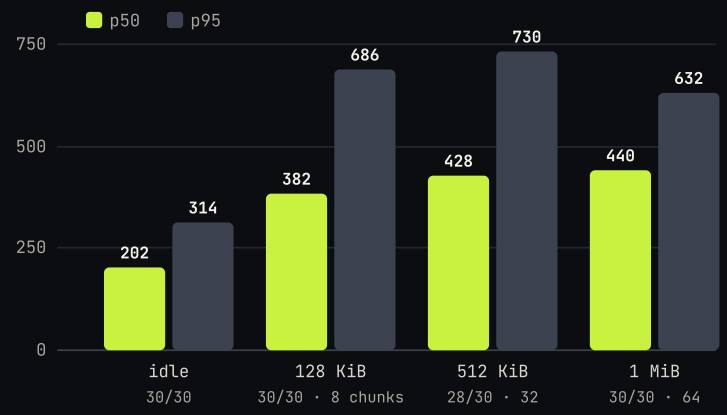
Scheduler: application-level chunking, receiver-confirmed backpressure, deterministic critical bypass.

04 THE PHYSICAL STRESS SESSION

342 attempts. Two hops.
The SOS never lost the race.

We drove three phones from three makers through text, offline Internet, SOS under bulk load, deliberate outages and relay crashes — automatically, for two hours, keeping every failure. An emergency packet had to cross a relay busy pushing up to a megabyte of photos. Every SOS that arrived, arrived first.

SOS ROUND TRIP VS QUEUED BULK DATA · 2 HOPS · MS



Eight times more queued data (8 → 64 chunks) moved the median SOS by only 58 ms. Priority pre-emption plus receiver-confirmed backpressure keep the emergency lane open at both hops.

88/88

DELIVERED SOS REACHED C BEFORE THE BULK TRANSFER FINISHED — AND THE BULK STILL ARRIVED INTACT (SHA-256)

30/30

AIRPLANE-MODE PHONE LOADED A WEB PAGE THROUGH TWO OTHER PHONES · P50 424 MS

0

DUPLICATE DELIVERIES ACROSS 251 TWO-HOP SUCCESSES

SURVIVES A DEAD RELAY

We killed the relay's app with a packet queued. On restart it reloaded the queue from its database and forwarded it — 9 of 10 times, median 21.7 s. A phone dying mid-crowd no longer loses the message.

SURVIVES AN ABSENT RECEIVER

Destination offline for 5–30 s: the relay held every packet and delivered all 40 when it returned. Store-and-forward now works across two hops, not one.

WHAT BROKE — AND WHY IT MATTERS

Cold discovery of the full chain took ~70 s (5/8 succeeded). A warm restart after hours of bulk traffic crashed C while loading its queue — out of memory at a 256 MiB heap, a bug only a long physical run could surface. Both are on the fix list.

> Every run was audited: matching relay receive/send logs, a two-hop route, reverse ACK evidence. Of 7 failures, 4 are MeshMind's and 3 the test harness's (ADB drop-outs, an operator stop) — all kept in the report, none replaced.

Session physical_20261006_01 (OnePlus 12R → Realme RMX5108 → Galaxy S24 Ultra, Android 16). Software-enforced A↔B↔C topology, one room; Nearby chose the bearer (Wi-Fi stayed on for B and C). Not RF-isolated, not Bluetooth-only, not event scale.

05 SAGE · SEMANTIC-AWARE GATEWAY & ENCOUNTER ROUTING

Flooding sends everything. *SAGE knows which copies matter.*

At every encounter a phone must decide: does giving this neighbour a copy actually help? We replayed real human movement and asked a time-travelling oracle for the true answer a million times — then trained a model that must answer using only the past.

1 · REAL CROWDS

11 public traces: conferences, a 69-day science exhibition, schools, offices, a hospital, 28 days of campus smartphone Bluetooth

2 · REPLAY

Synthetic MeshMind traffic, ACK gossip, shared airtime, finite buffers, gateways, churn

3 · ORACLE

Exact earliest-arrival over the future trace labels each decision — verified against exhaustive search

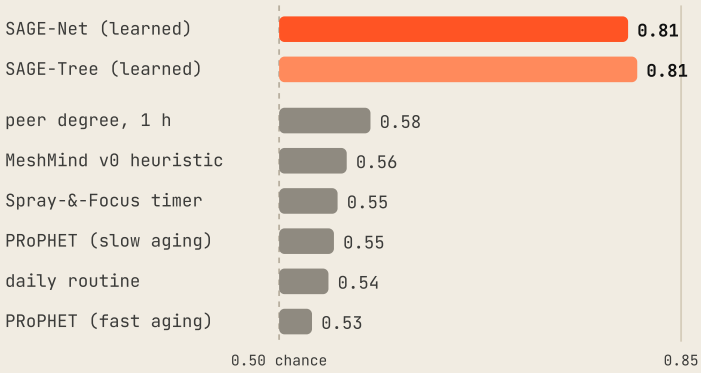
4 · LEARN

59 history-only features → 15k-parameter siamese network on one RTX 3050 in two minutes

5 · PROVE

Held-out events, people and days never seen in training or tuning

Spotting a copy that will change the outcome · held-out test, ROC AUC



WHY IT WORKS

The gain head sees both phones *and their relationship*. Two friends who walk together add little by copying to each other; a stranger heading for the exit adds a lot. Comparing the two phones' values alone scores only 0.59 — redundancy is the signal hand-written rules miss.

0.84

AUC: WILL THIS PHONE DELIVER? (PROPHET 0.67)

41–43%

OF USEFUL COPIES FOUND IN THE TOP 10% OF OFFERS (CROWD, CAMPUS)

ROUTING RESULT ON HELD-OUT EVENTS · THE HONEST VERSION

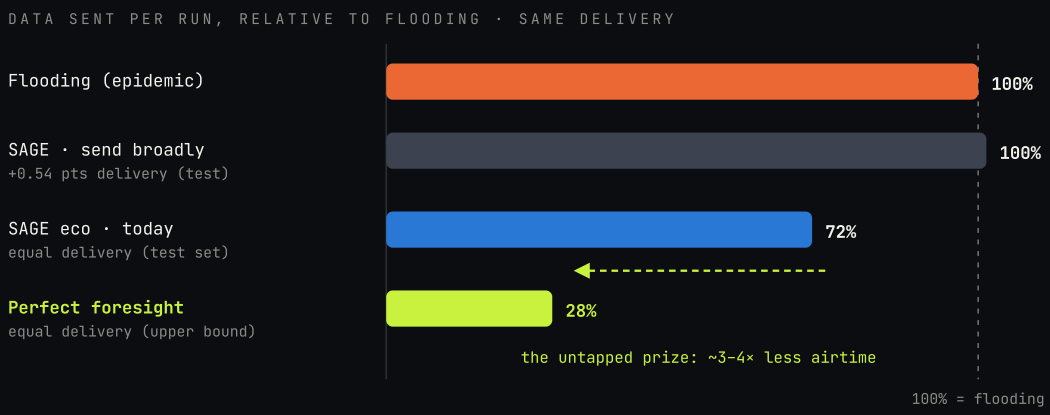
484 paired runs on 22 unseen episodes. **Max-delivery SAGE beat bounded flooding in all 17 stress scenarios: 40.8% vs 40.3% delivery** (+0.54 pts, 95% CI +0.44 to +0.65) at the same airtime, reaching 94.7% of the perfect-knowledge ceiling. **Eco SAGE matched flooding's delivery with 28% less airtime** (−0.11 pts, not significant), 28% fewer transmissions and a 191 s lower p95 latency. At low budgets it ties Spray-and-Focus. SOS outcomes were identical in every run. Gains are real but small; the airtime saving is the practical win.

AUC = probability a truly useful copy is ranked above a useless one; 0.5 is chance. Test split: SFHH conference, the last 6 days of the Copenhagen study, the last 14 days of the Infectious exhibition. CC BY-NC-SA / CC0 / MIT data licences apply.

05 WHAT PERFECT FORESIGHT WOULD DO

SAGE already routes like perfect foresight. *The next prize is a quarter of the battery.*

We swapped the model for a cheater that sees the future — it knows exactly which copies will matter. That is the best any routing model could ever do. The result tells us where the remaining value is: not in delivering more, but in delivering the same with far less radio time.



01 · DELIVERY: MAXED OUT

SAGE already matches foresight

Sending broadly, a perfect predictor beats flooding by +0.40 pts — and SAGE achieves exactly the same +0.40. Smarter models cannot deliver much more under these conditions.

02 · EFFICIENCY: WIDE OPEN

72% less data, same delivery

Copying only where it truly helps, perfect foresight matches flooding with 28% of the bytes — and under full buffers even delivers more (+0.6 pts), because useless copies stop crowding out useful ones.

03 · NEXT: CONFIDENT SKIPS

Learn when a copy is useless

SAGE eco captures 28% savings today. The gap is confidence at the selective end — so SAGE v2 targets "safe to skip" directly.

SAGE V2 RESEARCH PLAN

- Train on SAGE's own decisions (on-policy relabelling)
- Labels that account for packet loss and radio contention
- A calibrated "safe to skip" probability with conformal thresholds
- Calibrate airtime and contact length from phone telemetry

THE PITCH LINE

"SAGE already routes about as well as perfect foresight allows. Our next target is doing it with 3–4× less battery and airtime — the thing that decides whether a mesh survives a whole festival."

Headroom diagnostic: 162 runs on 6 validation Infectious days × {base, heavy load, queue saturation}; "perfect foresight" uses the offline oracle and ignores losses and contention, so real savings will be smaller. SAGE eco's 72% is the 484-run held-out test average.

06 AI THAT RUNS THE NETWORK ITSELF

A mesh that routes, schedules, *heals and saves itself.*

Today SAGE answers one question — should this neighbour get a copy? The roadmap turns it into the mesh's autonomic control plane: seven knobs, one tiny on-device policy, every knob beneath the same deterministic safety envelope.

#	CONTROL KNOB	WHAT THE AI DECIDES	STATUS
01	**Relay value**	Which neighbour should carry which packet — and whether the copy is redundant with friends who move together	PROVEN IN SIM
02	**Airtime scheduling**	Order of ordinary packets per encounter by expected delivery gain per byte; SOS is always first, deterministically	PROVEN IN SIM
03	**Value-aware buffers**	When storage is full, drop the copy least likely to matter — never an SOS	PROVEN IN SIM
04	**Gateway forecasting**	Which phone will still have Internet in ten minutes; route uplink traffic toward durable gateways, not flickering ones	NEXT
05	**Radio duty-cycling**	When to scan, advertise or sleep, and which bearer (BLE, Bluetooth, Wi-Fi Direct / Aware) — trading battery against expected encounters	NEXT
06	**Topology sculpting**	A phone holds only a handful of links. Keep the ones that maximise future reach; let redundant links go	RESEARCH
07	**Semantic transfer**	Description → thumbnail → preview → original, chosen by the predicted chance that the bigger version can arrive in time	VISION

TRAIN BIG

Thousands of replayed crowd-days on one laptop GPU; oracle labels from the future; leakage tests on every feature.

RUN SMALL

One ONNX graph, 75 KB, ~0.06 ms per decision on a laptop CPU. Android latency is measured before any claim.

SHIP SAFE

Shadow mode first: the phone logs what SAGE would do while the deterministic router stays in charge.

Federated SAGE

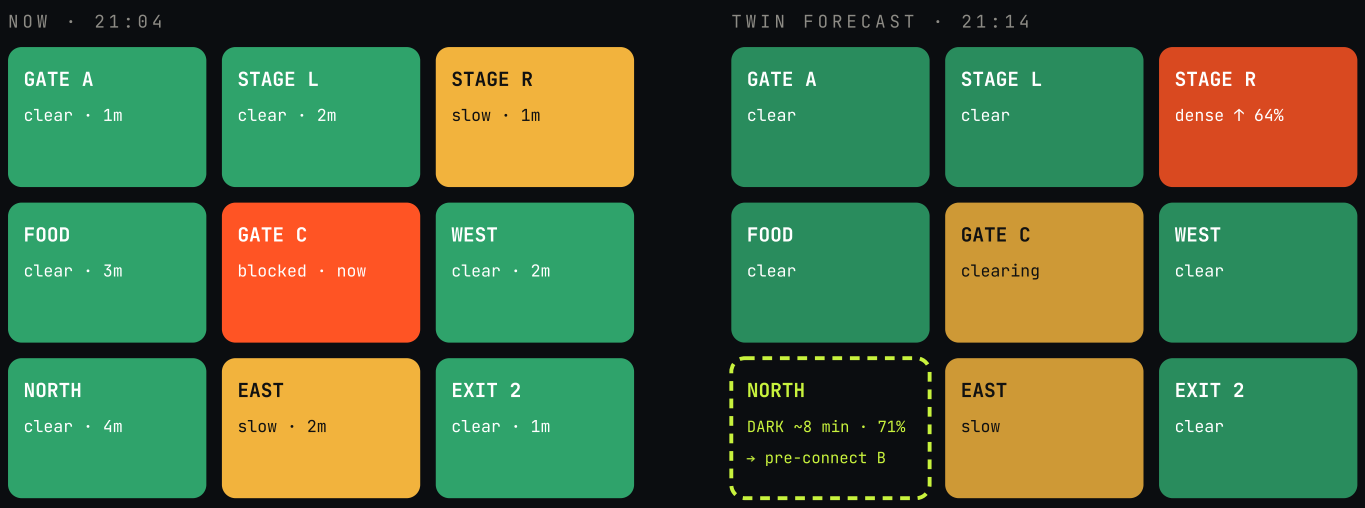
Every event teaches the next one. Phones compute model updates from consented telemetry; only aggregated weights leave the venue — never messages, contacts or locations. A model that has seen a hundred festivals is a moat no newcomer can download. VISION

Knobs 01–03 are implemented and evaluated in held-out simulation (research/sage/v1). Nothing here is wired into the Android router yet; integration starts in shadow mode.

07 SIGNATURE · MESH TWIN

See the network's future.
Fix the dead zone before it dies.

The organiser console keeps a live digital twin of the mesh: every phone's last-reported state, queue pressure, gateway health and contact history, replayed forward with the same simulator SAGE was trained in. It answers the question no dashboard can: where will we go blind in the next ten minutes?



01 · FORECAST

"North Stand: 71% chance of losing its last gateway route in ~8 min; 23 queued reports would strand."

02 · EXPLAIN

Shows why: two relay phones below 15% battery, crowd flow moving toward exits, one gateway flickering.

03 · PRE-CONNECT

Recommends a pre-approved checkpoint for one steward — or dropping a Relay Kit — before the zone goes dark.

04 · SCORE ITSELF

Every forecast is logged and graded afterwards. The twin publishes its own hit rate and false alarms.

WHAT MAKES IT POSSIBLE

The pieces already exist in the lab: a validated temporal oracle, a replay simulator, a leakage-tested history state and a learned reachability model (AUC 0.84 for "will this phone deliver?"). The twin points them at live telemetry instead of public traces.

HONESTY RULES

- Forecasts are shown as ranges with their own measured accuracy
- Stale data makes a zone grey, never green
- Humans approve every movement; recommendations that fail are logged

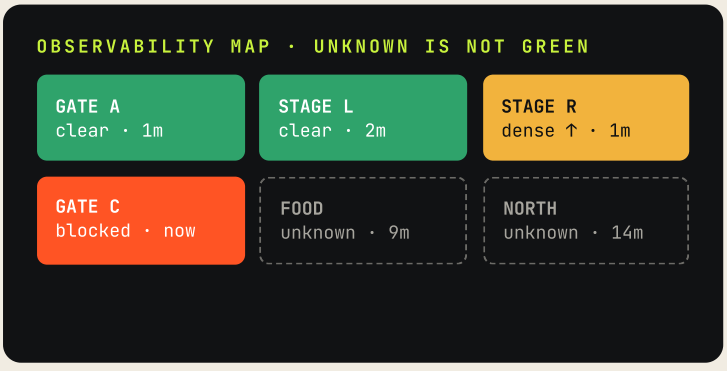
Status: RESEARCH → BUILDING. The reactive version (Reconnect a Zone) is in the 30-day plan; the predictive twin follows once field telemetry calibrates the simulator. Illustrative visual, not measured data.

08 CROWD PULSE

Every phone is a sensor.
The crowd is the instrument.

Cameras watch from fixed points. Twenty thousand phones are already inside the crowd, with radios, motion sensors, barometers and microphones. With consent, MeshMind turns them into a distributed instrument — computing features on the device, sending only tiny summaries, and showing uncertainty instead of verdicts.

SIGNAL	SENSOR	WHAT IT CAN HONESTLY TELL US	STATUS
Mesh density	BLE / Nearby scan counts	Relative density trend among *participating* phones; the same signal SAGE already learns from	NEXT
Motion coherence	Accelerometer + gyroscope	Whether nearby phones move together, stop-and-go, or turbulently — ambiguous alone (a dancing crowd also moves together)	RESEARCH
Fall + stillness	IMU impact + posture	A possible fall followed by no movement → opt-in Silent SOS candidate, confirmed or cancelled by the user	NEXT
Floor level	Barometer	Which level of a stand or building a report came from (relative pressure). Not body compression	NEXT
Loudness	Microphone, dB only	Sound-level trend per zone. Audio is never recorded, stored or transmitted	RESEARCH
Zone	Relay-Kit beacons + Wi-Fi	Coarse "which zone am I in" without GPS, from known checkpoint beacons	NEXT
Device health	Battery, thermal, screen	Which relays will die soon — direct input to SAGE and the Mesh Twin	BUILDING



PRIVACY CONTRACT

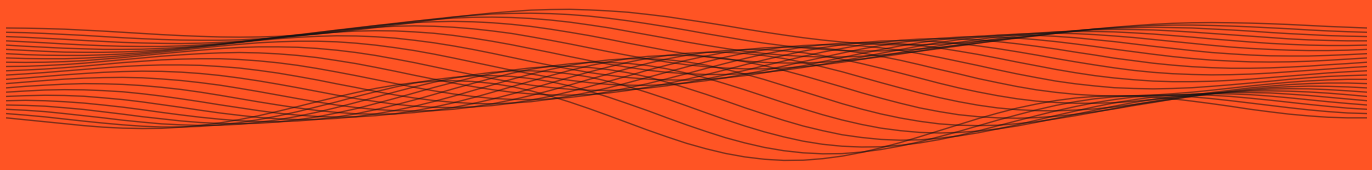
- Opt-in, visible, and only while Event Mode is on
- Features computed on the phone; raw sensor data never leaves it
- Rotating pseudonymous IDs; no names, numbers or precise tracks
- Aggregates per zone, with a minimum crowd size before display

Any operational safety claim requires calibration against video / UWB ground truth with published false-positive and false-negative rates. Crowd Pulse produces evidence for trained staff, not automatic instructions to the crowd.

09 — THE SIGNATURE LAYER

Most networks stop at delivered.

Ours stops at resolved — and sees what's next.



Six signatures turn a mesh into a response system. Three were in v2; three are new in v3 — and all six move an incident one step around the loop: observe, resolve, coordinate, verify.

S1 · BACKBONE

Living Incident

Survives partitions, shows conflicts

S2 · INTELLIGENCE

Ask the Right Person

One question that settles a decision

S3 · WOW MOMENT

Pre-connect a Zone

Move one phone before the zone goes dark

S4 · NEW

Mesh Twin

The network's ten-minute forecast

S5 · NEW

Silent SOS

Help without typing or speaking

S6 · NEW

Swarm Relays

Idle phones volunteer to heal the mesh

+ Attendee layer: Rendezvous Plan · Bhasha SOS · MeshReach · Last-Seen Trail. Research: SurgeWatch · Scenario Rehearsal.

S1 LIVING INCIDENT · THE BACKBONE

An incident that survives *a network split.*

Two people report a collapse near Gate C — one in Kannada, one in Hindi. MeshMind proposes that both describe one incident while keeping both originals. A medic accepts offline. The venue splits into two islands, then reconnects. Every report, assignment and state change merges back; nothing is silently overwritten.



MERGEABLE

Signed operation log with causal ordering; OR-set reports; multi-value assignee so conflicts are kept, not lost.

BLACK BOX

Hash-chained and replicated across gateways: a tamper-evident timeline of who knew what, and when.

SAFETY RULE

The basic SOS never waits for AI. AI can link, enrich or raise priority. It never suppresses, delays or merges away an explicit report.

S2 ASK THE RIGHT PERSON · THE VISIBLE INTELLIGENCE

The smartest output is often *a very good question.*

QUESTION → STEWARD R. · GATE C

Can people move freely through Gate C?

Clear | Slow | Blocked | Can't see

Triggered because Crowd Pulse saw stop-start motion near Gate C — congestion, dancing or normal waiting? One tap settles it.

- **Which uncertainty matters most** right now — value of information, not volume
- **Who** is placed, authorised and fresh enough to answer
- A **question budget** per staff member: no alert fatigue
- **Approved templates only** — the model chooses, never free-writes instructions
- Answers **decay**: a ten-minute-old "clear" turns the zone back to unknown
- Rules-based selector ships first; **a learned selector must beat it to ship**

Status: BUILDING (30-day plan). Proof: the record survives a defined restart/reconnect suite; targeted questions resolve a staged incident faster with fewer notifications than broadcasting.

S3 PRE-CONNECT A ZONE · THE MEMORABLE MOMENT

Move one phone.
The event never goes blind.

v2 reacted: when a zone stopped acknowledging, MeshMind recommended where a steward could stand to restore contact. v3 anticipates: the Mesh Twin flags the zone minutes before it drops, and the organiser can approve the move while the backlog is still small.

01 · PREDICT

Twin forecasts loss of the zone's last durable route.

02 · SCORE

Rank pre-approved checkpoints by recently measured link quality — not RSSI-guessed positions.

03 · APPROVE

Organiser approves; steward confirms the route is clear and accessible.

04 · VERIFY

Backlog delivers; zone marked known; the outcome is logged — including failures.

RELAY KIT

A spare Android phone on a power bank, clipped at a checkpoint, turns a recommendation into a permanent fix. Pre-place a handful at known dead spots before doors open. They double as zone beacons for Crowd Pulse.

SKY RELAY · VISION

For melas and floods: tethered-balloon or mast-mounted relay pods that lift a Relay Kit above the crowd's body-blocking, extending range across ghats and river banks. Pilot only with permits and a safety review.

S6 SWARM RELAYS · NEW IN V3

Idle phones *volunteer to heal the mesh.*

When SAGE sees a bridge forming — a phone with high battery, a charger, and contact with two otherwise separate groups — it can ask that attendee, once, to keep MeshMind awake as a relay for the next 30 minutes. Consent is explicit, the battery cost is shown up front, and the reward is visible: "you carried 41 messages tonight, including 2 SOS". Staff phones are always relays; attendee relays are a gift, never an assumption.

- Asked only where the twin predicts real value
- Battery floor and time limit chosen by the user
- Relay credits: festival perks, recognition, priority rendezvous
- Never extends what a relay can read: relays forward ciphertext

Status: S3 reactive version BUILDING; predictive pre-connect RESEARCH; Swarm Relays NEXT; Sky Relay VISION. Human approval always precedes physical movement.

S5 SILENT SOS + BHASHA SOS

No typing. No speaking. *Still heard.*

In a crush you cannot raise your arms; in panic you cannot type; in a crowd of twelve languages the medic may not read yours. MeshMind gives the SOS three entrances — and every one of them leaves on the deterministic critical path within milliseconds, before any AI enrichment.

ENTRANCE 1 · PATTERN

Press, don't look

Five presses of the power button or a held volume-key pattern sends a basic SOS with zone and time. Works with the screen off, in a pocket.

ENTRANCE 2 · BODY

Fall + stillness

Opt-in: an impact followed by no movement starts a 20-second countdown the user can cancel; otherwise an SOS leaves with the motion evidence attached.

ENTRANCE 3 · VOICE

Bhasha SOS

Speak in Kannada; the medic reads Hindi. On-device transcription → a tiny structured packet that renders in the responder's language.

```
"ಅಮ್ಮ ಬಿದ್ದಿದ್ದಾರೆ, ಉಸಿರಾಡ್ತಿಲ್ಲ"  →  { type: MEDICAL, sev: CRITICAL, people: 1,
                                  sym: [collapse, NOT breathing], zone: Gate C,
                                  floor: +1 (barometer), conf: 0.82 }  →  medic (hi)
```

Built for negation

"Not breathing", "not injured" and "no longer blocked" must stay distinguishable. The evaluation set is built around negation, code-switching and vague locations; native speakers verify; real recordings and speakers stay separate between training and evaluation.

Order of operations

1. Basic SOS leaves instantly — deterministic, CRITICAL
2. Enrichment packets follow: category, severity, floor, language
3. AI may **raise** priority or link reports; it never lowers, delays or merges away an explicit SOS
4. False-trigger rate is measured and published per entrance

> Accessibility is safety: the same entrances serve people who are deaf, non-verbal, injured, or simply holding a child with both hands.

Status: Bhasha SOS KEEP (two tested languages first). Pattern SOS NEXT. Fall + stillness RESEARCH → NEXT after false-alarm measurement on target phones.

10 WHY ATTENDEES JOIN

Safety is the reason to exist.
Everyday usefulness is the reason to join.

A1 Rendezvous Plan — a meeting plan that still works after friends lose contact

The group picks a meeting point and a fallback time; each member acknowledges it. When someone proposes a change, everyone sees exactly who has acknowledged the new version. "We suggested a new meeting point" is never shown as "everyone agreed to it".

`Versioned plans` `Per-member ACK` `Works fully offline`

A2 MeshReach — family can reach you when your phone has no Internet

A message from outside enters through a consenting gateway and crosses the mesh to one specific attendee. Private addressing via rotating reachability tokens, multiple candidate gateways for critical traffic, expiry, and receipts back to the sender. SAGE's gateway forecasting chooses the gateways most likely to stay up.

`Rotating tokens` `Receipts` `Gateway forecasting`

A3 Last-Seen Trail — find a lost child without a single camera

Mesh encounters already say which relays and zones a phone passed. With a guardian token set up at entry, a guardian (and only a guardian, with staff) can see the child's last-seen zones and times — "Ghat 4, 12 minutes ago, via Relay Kit 17". Rotating IDs keep everyone else anonymous; trails expire after the event. Built for melas, where families are separated by the thousand.

`Guardian-only` `Expires` `No cameras, no faces` `VISION`

A4 Verified alerts — the organiser speaks, the mesh carries it, nobody can fake it

Signed with event authority keys and rendered in the attendee's language. Fake "stampede!" messages are the most dangerous rumours in a crowd; unsigned alerts are visibly marked as unverified and never get the critical path.

`Ed25519 authority keys` `Multilingual` `Rumour-resistant`

ESCAPE PATHS FROM THE LOCAL MESH

Internet gateway — validated, including two hops away. **SMS gateway** — conditional; Play SMS-permission policy needs review.

Satellite-capable phones as gateways — vision: as satellite messaging reaches mainstream Android phones, any such phone in the crowd could become the event's last-resort uplink, with SAGE routing only the most critical traffic to it.

Attendee participation is opt-in, permissioned and only counts while Event Mode is active. The staff network must be useful without it.

11 THE AMBITIOUS BETS

The moonshots — *with the maths stated up front.*

SurgeWatch RESEARCH HYPOTHESIS

Could correlated motion anomalies at the front of a crowd give people further back useful lead time? Only if this inequality holds:

$$T_{wave} > T_{detect} + T_{corroborate} + T_{deliver} + T_{respond}$$

We have measured $T_{delivery}$ on phones over two hops: SOS p50 202 ms idle, 440 ms through a 1 MiB transfer. Crowd-disaster research supports studying stop-and-go and turbulent motion; it does not establish a warning lead time. So we measure each term separately, publish it, and never schedule "crush prediction" as a feature.

Scenario Rehearsal SIMULATION

"Gate C is reported blocked. Compare: leave instructions unchanged, ask staff to inspect Gate B, or use a pre-approved diversion." The tool evaluates options against a verified venue map, uncertain crowd estimates, route capacities, accessibility and evidence freshness — and says when the evidence is too thin to choose.

THE COMPUTE PROBLEM NOBODY MODELS

Couple crowd simulation to communication simulation. An instruction changes how people move; movement changes the mesh; the mesh changes who receives the next instruction. Our SAGE replay engine is the communication half of that loop.

MOONSHOT	QUESTION WE WOULD ANSWER	FIRST HONEST MILESTONE
Federated SAGE	Can routing models improve across events without any raw data leaving phones?	Two-event federated round in simulation
Predictive Mesh Twin	How accurately can we forecast a zone losing contact 5–10 minutes ahead?	Published hit rate on a staged event
Topology sculpting	With only a handful of links per phone, which links maximise reach for the whole venue?	Sim gain vs Nearby's default selection
Learned radio duty cycle	How many hours of Event Mode can learned scanning add at equal delivery?	Battery A/B on three phone models
Crowd–comms co-simulation	Which instructions actually reach people in time to change movement?	JuPedSim × SAGE replay prototype

Demo SurgeWatch only with recorded motion traces or separated volunteers making ordinary movements — never a staged shove. Any operational claim needs calibration against video / UWB ground truth.

12 SAFETY BY CONSTRUCTION

AI proposes. *Deterministic rules dispose.*

The more ambitious the AI, the harder the safety boundary has to be. MeshMind's boundary is not a policy document — it is code, and it is tested with a hostile model.

THE ADVERSARIAL TEST

We plug in a model that refuses every copy and crashes if it is ever shown an SOS. With 100% critical traffic its delivery, latency and transmissions are **identical** to flooding; in mixed traffic it scores thousands of ordinary rows and never sees one SOS. If the model is missing or corrupt, the hand-written heuristic takes over.

```
decide(packet, peer):
  if packet.priority == CRITICAL:
      return REPLICATE        # before any model
  if peer is destination:
      return FORWARD
  try:   score = sage(features)   # ordinary only
  except: score = heuristic()     # safe fallback
  # simulator re-applies the CRITICAL gate
  # AFTER the policy returns
```

RULE	HOW IT IS ENFORCED
SOS never waits for AI	Critical gate precedes model access; scheduler sends CRITICAL first; buffers never evict CRITICAL for ordinary traffic
No future leakage	History state rejects time travel; a test rewrites the future of a trace and proves every earlier feature is unchanged
Shadow before authority	On phones, SAGE first logs recommendations while the deterministic router decides; promotion needs a held-out win
Humans move, models suggest	Every physical action — a steward, a Relay Kit — needs organiser approval and is logged with its outcome
Unknown is not green	Freshness decays every observation; missing evidence renders grey on every map
Relays can't read	End-to-end encrypted payloads; interpretation happens only on the origin or an authorised recipient
If ML loses, ship the baseline	Every learned component has a non-learning baseline that it must beat on held-out data to ship

> Known gap we are closing before integration: the Android router's optional NextHopAdvisor hook currently runs for CRITICAL packets too. It must gain an explicit critical bypass and a test before SAGE is attached.

Safety tests live in research/sage/tests (adversarial model, fallback, eviction, leakage) and the core router's JVM test suite.

13 SYSTEM ARCHITECTURE

A local mesh data plane.
An intelligence plane that learns.

CLOUD (OPTIONAL)

Family ingress · reachability tokens · federated SAGE aggregation · cross-event evaluation

ORGANISER CONSOLE (LOCAL)

Incident fusion · Mesh Twin · question selection · observability map · checkpoint scoring

INCIDENT LAYER

Signed op-log · causal merge · conflict surfacing · hash chain

INTELLIGENCE PLANE · ON PHONE

SAGE ONNX policy (relay · schedule · evict) · Crowd Pulse features · Silent SOS detectors · report structuring · abstention

ROUTING

IDs · TTL · dedupe · ACK/SUMMARY gossip · store-and-forward · bounded replication · backpressure · critical bypass

TRANSPORT ABSTRACTION

Nearby Connections · simulation · fault injection · future Wi-Fi Aware / BLE

PHONE RADIOS + SENSORS

Bluetooth · BLE · Wi-Fi · IMU · barometer · microphone (dB only)

LAYER	STACK
Android app	Kotlin 2 · Jetpack Compose · coroutines · Room
Event Mode	Foreground service (connectedDevice) · explicit, tested early
Transport	Nearby Connections P2P_CLUSTER behind a transport interface
Router core	Pure Kotlin/JVM; same code in phones and simulation
SAGE runtime	ONNX Runtime Mobile or LiteRT · 59 versioned features · 15k params · shadow mode first
SAGE lab	Python 3.11 · PyTorch (CUDA) · scikit-learn · temporal oracle · replay simulator · 11 public traces
Crypto	Tink / libsodium · Ed25519 · X25519 + XChaCha20-Poly1305 · Android Keystore
Incident layer	Lamport / vector clocks · OR-set · multi-value register · SHA-256 chain
On-device AI	whisper.cpp or offline SpeechRecognizer · LiteRT classifiers · SensorManager features
Console	Ktor or FastAPI · React + Vite · MapLibre offline tiles · Mesh Twin on the SAGE simulator
Quality	JUnit · 25 router tests · 100+ research tests · 3-phone ADB benchmark harness · CSV evidence

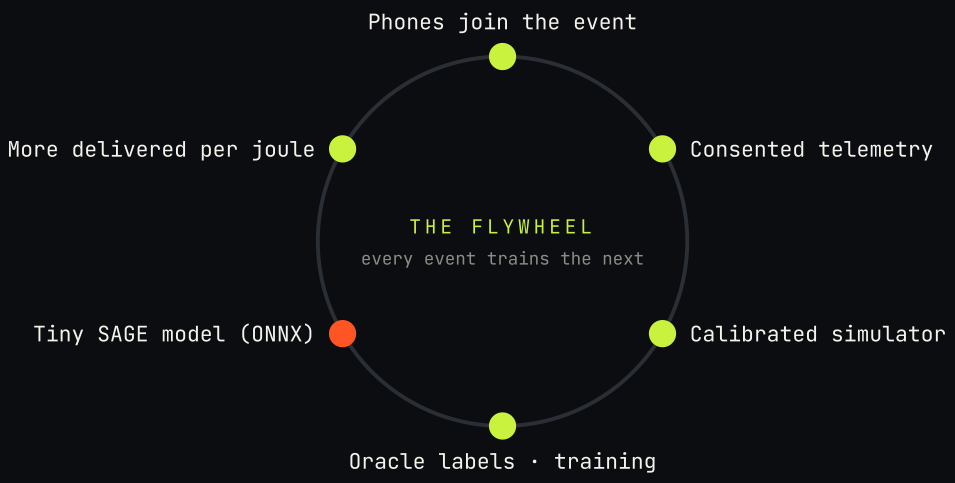
ENVELOPE

```
event_id · msg_id · dest · created · expires
priority · hop_budget · replicas · sig
payload (E2E-encrypted when needed)
```

A 20,000-person deployment is not 20,000 simultaneous radio links. Each phone keeps a handful of links; application-layer routing spans the venue — which is exactly why choosing those links and copies well matters.

14 THE LEARNING FLYWHEEL

Big compute in the lab.
Small, dependable models on the phone.



PROGRAMME 1

Failure laboratory

Replay real routing code against churn, gateway loss, uneven adoption, killed processes, bursty traffic. Model radio contention — perfect-link simulators lie.

PROGRAMME 2

SAGE vs strong baselines

Epidemic, Spray-and-Wait, Spray-and-Focus, PRoPHET and a same-mechanics PRoPHET-value policy. Tuned on validation, reported on held-out events.

PROGRAMME 3

Hard language set

Noise, code-switching, vague locations, negation. Native speakers verify; speakers never shared across train and test.

PROGRAMME 4

Distil everything

Expensive simulations become tiny models: which relay stays useful, which question matters, which reports are one incident — and when to abstain.

V2 TARGET → V1 RESULT (HELD-OUT SIMULATION)

Target: −25% bytes at equal delivery, emergency delivery and p95 unchanged. **Result: −28% total airtime at statistically equal delivery; SOS delivery and latency identical in all 484 runs. Not yet met: duplicate bytes fell only 12%.** Next target: 3–4× less airtime at equal delivery (perfect-foresight bound: −72%), then the same on phones.

WHAT WE COLLECT · AND NEVER COLLECT

Consented, pseudonymous field telemetry: peer counts, contact durations, queue length, battery, packet size/type/TTL, relay choice, outcome, latency, hops. **Never** message text, media, phone numbers or persistent precise location.

Public training data used today: SocioPatterns (conference, exhibition, school, workplace, hospital) and the Copenhagen Networks Study. CRAWDAD traces require licence steps and are not used until obtained properly.

15 WHERE OTHERS STOP

Everyone built a piece.
We're building the nervous system.

DIMENSION	HOW OTHERS TYPICALLY WORK	WHERE IT FAILS AT AN EVENT	MESHMIND
Finish line	Message delivered	Nobody knows if anyone took responsibility	**Accepted → on scene → resolved,** confirmed to the reporter
Routing	Flood, or fixed rules	Battery and airtime spent on redundant copies	**Learned relay value:** AUC 0.81 vs 0.53–0.58 for rules
Network split	Last write wins	Two medics dispatched; one silently erased	**Mergeable incidents;** conflicts shown to a human
Blind spots	No reports looks like all clear	Silent zones are the dangerous ones	**Observability map + Mesh Twin forecast**
Under load	FIFO, trust the transport's "sent"	SOS stuck behind megabytes (2,374 ms)	**Receiver-confirmed backpressure: 145 ms**
SOS input	Typed text	A crushed or panicking person cannot type	**Silent SOS + Bhasha SOS**
Sensing	Fixed cameras	Blind spots; faces; privacy	**Crowd Pulse:** consented on-device evidence
Evidence	Feature lists and demos	Organisers can't compare reliability	**Published benchmarks with ceilings and baselines**

OUR HONEST UNIQUENESS CLAIM

We cannot promise no one has built any single piece. We will show that MeshMind gets an incident from "someone needs help" to "a responder has accepted it, and the reporter knows" — when connectivity is failing — better, under measured conditions.

PLAYER	WHAT'S MISSING FOR EVENT RESPONSE
bitchat	Privacy messenger; no incident workflow, roles or observability
Bridgefy	Messaging and broadcast; no responsibility tracking
Meshtastic · goTenna	Extra hardware; not event operations
APPIX · PixMob	One-way alerts and lighting; no two-way loop
CCTV analytics	Fixed infrastructure; no staff or attendee comms
Radios + WhatsApp	Unstructured; dies with cellular; nothing to audit

"How others typically work" describes common primary designs, not a claim that no product has any adjacent capability. Descriptions reflect each player's primary published purpose.

16 START WITH THE TEAM · EXTEND TO THE CROWD · EARN THE CITY

From a college fest *to the Kumbh.*

NOW

3-phone multi-hop; SAGE v1 proven in sim; benchmark harness

DAY 30

Incident loop, Ask the Right Person, Reconnect a Zone, SAGE in shadow mode

DAY 90

First 15–30-staff drill at a college fest; SAGE v2 targeting 3–4× less airtime; benchmark report v1 public

MONTH 6

First paid event; Silent SOS; Mesh Twin beta; SAGE ships or is dropped on evidence; iOS interop

MONTH 12

Stadium + ticketing SDK beta; federated SAGE across events; Crowd Pulse calibration study

MONTH 24

Mela-scale pilot with authorities (e.g. Simhastha 2027 Nashik, if invited); civic mesh for floods

Revenue lines, in order

Per-event service	Staff app, console, Relay Kits, drill report
Annual venue licence	Campuses, stadiums, festival operators
Enterprise SDK	Attendee reach inside ticketing / event apps
Government	Melas, pilgrimages, disaster response — later
Insurers · telcos	Black Box audit trails; tower offload — later
Data products	Anonymised benchmark + drill reports for licensing bodies

How we validate the first hypothesis

Supervised drills alongside an organiser's existing radio / WhatsApp procedure, measured side by side:

- Time from report to responder acceptance
- Missed and duplicated dispatches
- Staff workload and notifications per incident
- Airtime and battery per delivered message (SAGE vs flooding)

RELIABILITY

Measured on mixed Android phones, screens off, in pockets

WORKFLOWS

Incident processes validated with real teams and drills

DATA

Consented telemetry and evaluation sets nobody else has

MODELS

SAGE trained on every event we run — federated, never raw

THE 24-MONTH MOONSHOT · A TARGET, NOT A CLAIM

A mela where no separated family waits more than an hour, no zone goes dark without someone seeing it coming, and every SOS reaches a responder who accepts it — carried by phones the pilgrims already own.

Roadmap dates are proposed targets, not claims. Mela and government milestones depend on invitations, permits and safety review.

17 HACKATHON DEMO · 5 MINUTES, ONE STORY

A disconnected incident becomes *understood, accepted — and foreseen.*

0:00–0:30	**Set the stage.** A venue map, several real phones, Internet off on attendee devices. Conditions visible on screen.	LIVE
0:30–1:10	**Silent SOS under load.** A phone in a pocket: five presses. The SOS cuts through a 1 MB transfer across two hops while the bulk transfer keeps running.	LIVE
1:10–1:50	**Bhasha + Ask the Right Person.** A spoken Kannada report links to the SOS; a steward gets the one missing question and taps "Blocked".	LIVE
1:50–2:30	**The twin sees it coming.** Mesh Twin flags North Stand: its last relay's battery is falling and its gateway is flickering.	SIMULATION-BACKED
2:30–3:20	**Pre-connect a Zone.** The organiser approves a checkpoint; a steward walks there. When the relay is pulled, the zone never goes dark.	WOW MOMENT
3:20–4:00	**Close the loop.** A medic accepts; the reporter sees "help is on the way"; a judge's own 4G phone receives the update through a gateway.	LIVE
4:00–5:00	**Prove the brain.** Replay the same night with flooding vs SAGE: airtime saved, delivery kept, SOS untouched — next to the perfect-knowledge ceiling.	LABELLED SIM

30-SECOND CLOSE

"Every mesh app stops at 'delivered'. MeshMind is a nervous system for crowds: the network routes itself with a model that learned from a million real decisions, the phones sense what cameras can't, and the response team sees where the event is about to go blind — before it does. We've already sent an SOS through a megabyte transfer in 145 milliseconds, and across two hops ahead of a megabyte of bulk data 88 times out of 88 on three phones. Now we're closing the loop."

MEASURED LIVE ON STAGE

SOS latency through bulk traffic, hop path and acceptance time — shown as they happen.

SHOWN WITH ITS CEILING

SAGE replay results always appear next to flooding and the perfect-knowledge ceiling.

LABELLED, ALWAYS

Every simulated or fault-injected element carries a visible label on screen.

Software-disabled links are labelled as fault injection. "True radio multi-hop" is reserved for a test with established physical RF topology.

A CLAIMS DISCIPLINE

What we can say today — *and what we can't yet.*

CLAIM	WHAT WE CAN SAY NOW	STATUS
Offline peer communication	Validated on three physical Android phones from three makers	VALIDATED
Store-and-forward · SOS pre-emption	Validated; 145 ms during a 1 MB transfer via chunking + backpressure	VALIDATED
Application multi-hop A→B→C	2-hour session: 251/258 two-hop deliveries, 0 duplicates; SOS 88/88 ahead of bulk; offline web 30/30; software-enforced topology	VALIDATED · SW TOPOLOGY
Physical RF-isolated multi-hop	Needs a test with established physical topology	NOT YET
Fast cold discovery	Measured weakness: 5/8, p50 70 s to form a 3-phone chain; warm-restart OOM found	FIXING
SAGE identifies useful copies	AUC 0.81 vs 0.53–0.58 for hand-written rules on held-out events (simulation on real traces)	PROVEN IN SIM
SAGE routing advantage	Held-out sim: +0.54 pts delivery vs flooding at equal airtime (all 17 scenarios), or equal delivery with −28% airtime; ties Spray-and-Focus at low budgets; untested on phones	SIM · NOT ON PHONES
SAGE efficiency headroom	Perfect foresight matches flooding with 28% of the bytes; SAGE eco reaches 72% today — the v2 research target	RESEARCH TARGET
Living Incident · questions · pre-connect	Designed; 30-day build plan with exit criteria	BUILDING
Mesh Twin · Silent SOS · Swarm Relays	Designed; components exist in the lab or as Android sensor APIs	NEXT / RESEARCH
SurgeWatch / crowd sensing	Research hypothesis; no warning lead time claimed	RESEARCH
Last-Seen Trail · Sky Relay · satellite gateways	Vision; privacy, permit and hardware work required	VISION
Large-crowd scale · emergency use	Needs staged field evidence · not a certified emergency system	NOT YET · NOT CERTIFIED

MESHMIND

When the network fails,
the crowd thinks for itself.

THE ASK

A pilot event · 15–30 staff phones
a response team willing to drill with us
a venue with a known dead zone

Concept brochure v3, October 2026 — "the ambition edition". Phone figures from the October 2026 two- and three-phone sessions; SAGE figures from research/sage v1 held-out evaluation. Vision items are labelled and are not claims.